

## Assignment 2: Use case diagram

### Exercise 1: sales process

In a store, the sales process is as follows:

The customer enters, goes through the shelves, possibly requests information or carries out tests, takes items (if there is sufficient stock), goes to the checkout where he pays for his purchases (with any accepted payment method). He may possibly benefit from a reduction.

Model this situation with a use case diagram

### Exercise 2: Library System

Create a use case diagram for a library system that allows users to search for books, borrow books, return books, and reserve books.

Consider different user roles like librarian, member, and guest. Model this situation with a use case diagram

### Exercise 3: Online Shopping Platform

Create a use case diagram for an online shopping platform that allows users to browse products, add items to the cart, proceed to checkout, and make payments.

### Exercise 4 – Course registration

The following should be textually analysed and a use case diagram created containing several use cases. Identify the actors, use cases and associations.

At the start of each semester a student can request a prospectus containing a course list. Information about a course is provided, such as the tutor, department and pre-requisites.

The new system will allow students to create a schedule, then select four courses. Each student chooses two others in case their first choices become full or are cancelled. No course can have more than 10 students. No course can have less than 3 students or it will be cancelled.

This will be the same functionality as available to other internal users of the system.

When registration is complete, the registration system sends a message to the billing system to send out a bill to the student.

Tutors use the system to and which classes they are teaching and who the students are. The registrar will administer the system.

For a period at the beginning of the semester the student can change their schedule. Students must be allowed to access the system during this time to add or delete courses.

Model this situation with a use case diagram